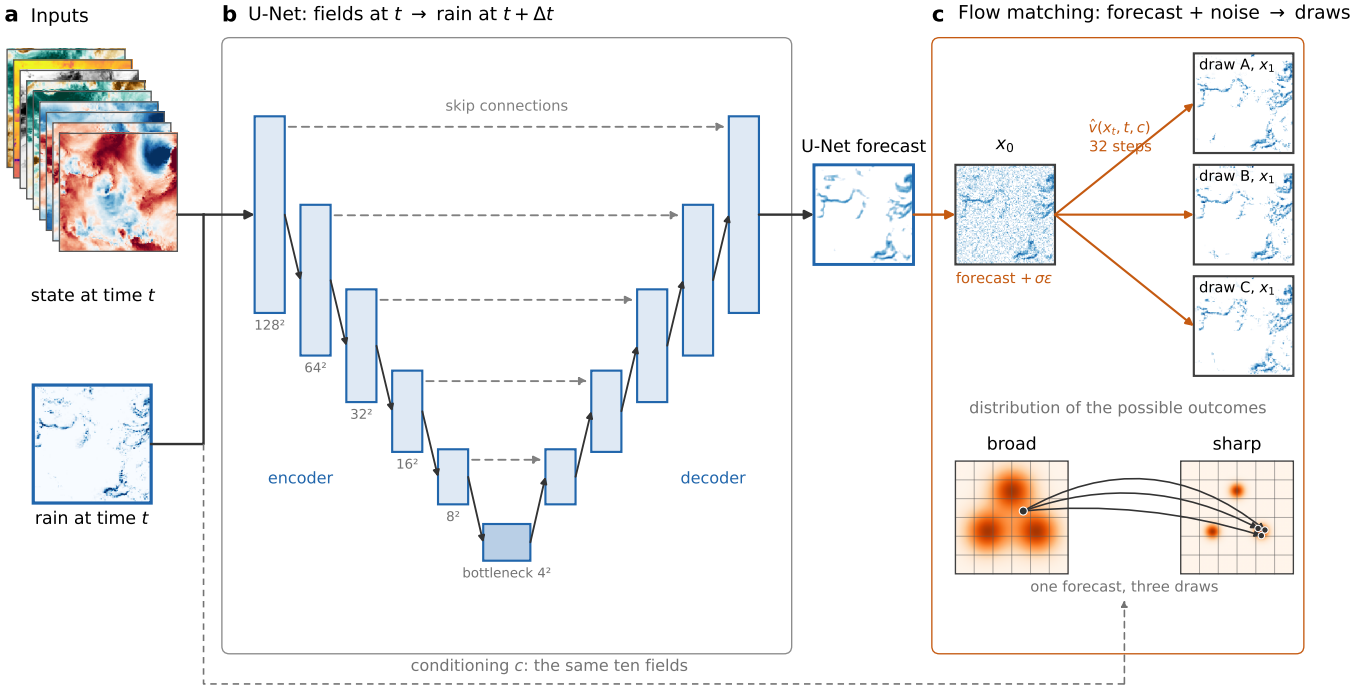

a Inputs
state at time t
rain at time t
b U-Net: fields at t → rain at t + Δt
skip connections
128²
64²
32²
16²
8²
bottleneck 4²
encoder
decoder
conditioning c: the same ten fields
U-Net forecast
c Flow matching: forecast + noise → draws
x0
forecast + σε
v̂(x_t, t, c)
32 steps
draw A, x1
draw B, x1
draw C, x1
distribution of the possible outcomes
broad
sharp
one forecast, three draws